

types of research and illustrates that exploratory research is conducted during the early stages of decision making. At this point, the decision situation is usually highly ambiguous and management is very uncertain about what actions should, or even could, be taken. When management is aware of the problem but lacks some knowledge, descriptive research is usually conducted. Causal research requires sharply defined problems.

Each type of research also produces a different type of result. In many ways, exploratory research is the most productive since it should yield large numbers of ideas. It is part of the “domain of discovery,” and as such, unstructured approaches can be very successful. Too much structure in this type of research may lead to more narrowly focused types of responses that could stifle creativity. Thus, although it is productive, exploratory research results usually need further testing and evaluation before they can be made actionable. At times, however, managers do take action based only on exploratory research results. Sometimes, management may not be able or may not care to invest the time and resources needed to conduct further research. Decisions made based only on exploratory research can be more risky, since exploratory research does not test ideas among a scientific sample.⁸ For instance, a business school professor may ask a class of current MBA students for ideas about an online program. Although the students may provide many ideas that sound very good, even the best of them has not been tested on a sample of potential online MBA students.

Descriptive research is typically focused around one or more fairly specific research questions. It is usually much more structured and, for many common types of business research, can yield managerially actionable results. For example, descriptive research is often used to profile a customer segment both demographically and psychographically. Results like this can greatly assist firms in deciding when and where to offer their goods or services for sale.

Causal research is usually very focused around a small number of research hypotheses. Experimental methods require tight control of research procedures. Thus, causal research is highly structured to produce specific results. Causal research results are often managerially actionable since they suggest that if management changes the value of a “cause,” some desirable effect will come about. So, by changing the training program, the cause, an increase in employee productivity, can result.

Stages in the Research Process



Business research, like other forms of scientific inquiry, involves a sequence of highly interrelated activities. The stages of the research process overlap continuously, and it is clearly an oversimplification to state that every research project has exactly the same ordered sequence of activities. Nevertheless, business research often follows a general pattern. We offer the following research business stages:

1. Defining the research objectives
2. Planning a research design
3. Planning a sample
4. Collecting the data
5. Analyzing the data
6. Formulating the conclusions and preparing the report

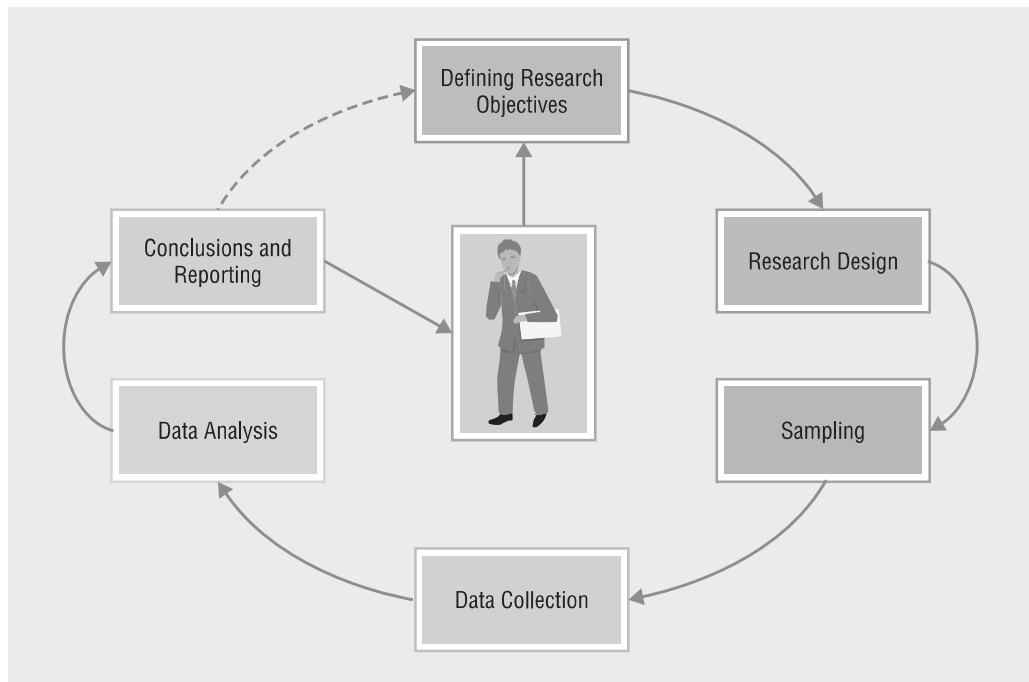
Exhibit 4.5 on the next page portrays these six stages as a cyclical or circular-flow process. The circular-flow concept is used because conclusions from research studies can generate new ideas and knowledge that can lead to further investigation. Thus, there is a dashed connection between *conclusions and reporting* and *defining the research objectives*. Notice also that management is in the center of the process. The research objectives cannot be properly defined without managerial input. After all, it is the manager who ultimately has to make the decision. It is also the manager who may ask for additional research once a report is given.

In practice, the stages overlap somewhat from a timing perspective. Later stages sometimes can be completed before earlier ones. The terms *forward linkage* and *backward linkage* reflect the interrelationships between stages. **Forward linkage** implies that the earlier stages influence the later stages. Thus, the research objectives outlined in the first stage affect the sample selection and

forward linkage

Implies that the earlier stages of the research process influence the later stages.

EXHIBIT 4.5

Stages of the Research Process**backward linkage**

Implies that later steps influence earlier stages of the research process.

Research is sometimes directly actionable. The results may also suggest ideas for new studies.

the way data are collected. The sample selection question affects the wording of questionnaire items. For example, if the research concentrates on respondents with low educational levels, the questionnaire wording will be simpler than if the respondents were college graduates.

Backward linkage implies that later steps influence earlier stages of the research process. If it is known that the data will be collected via e-mail, then the sampling must include those with e-mail access. A very important example of backward linkage is the knowledge that the executives who will read the research report are looking for specific information. The professional researcher anticipates executives' needs for information throughout the planning process, particularly during the analysis and reporting.

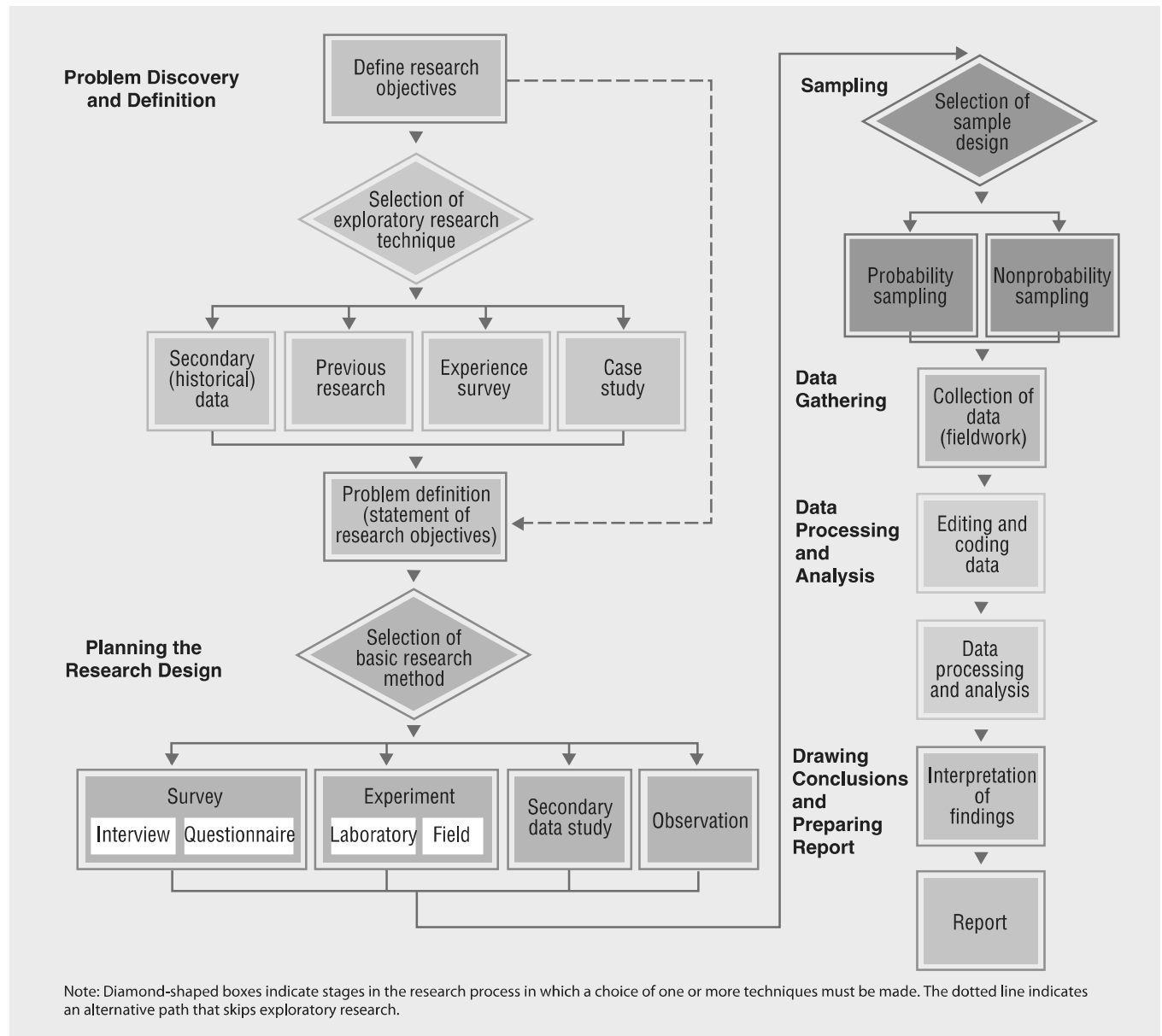
Alternatives in the Research Process

The researcher must choose among a number of alternatives during each stage of the research process. The research process can be compared to a map. It is important to remember that there is no single "right" path for all journeys. The road one takes depends on where one wants to go and the resources (money, time, labor, and so on) available for the trip. The map analogy is useful for the business researcher because there are several paths that can be followed at each stage. When there are severe time constraints, the quickest path may be most appropriate. When money and human resources are plentiful, more options are available and the appropriate path may be quite different.

Chapter 1 introduced the research process. Here, we briefly describe the six stages of the research process. Later, each stage is discussed in greater depth. Exhibit 4.6 shows the decisions that researchers must make in each stage. This discussion of the research process begins with research objectives, because most research projects are initiated to remedy managers' uncertainty about some aspect of the firm's business program.



EXHIBIT 4.6 Flowchart of the Business Research Process



Defining the Research Objectives

Exhibit 4.6 shows that the research process begins with **research objectives**. Research objectives are the goals to be achieved by conducting research. In consulting, the term **deliverables** is often used to describe the objectives to a research client. The genesis of the research objectives lies in the type of decision situation faced. The objectives may involve exploring the possibilities of entering a new market. Alternatively, they may involve testing the effect of some policy change on employee job satisfaction. Different types of objectives lead to different types of research designs.

In applied business research, the objectives cannot really be determined until there is a clear understanding of the managerial decision to be made. This understanding must be shared between the actual decision maker and the lead researcher. We often describe this understanding as a problem statement. In general usage, the word *problem* suggests that something has gone wrong. This isn't always the case before research gets started. Actually, the research objective may be to simply clarify a situation, define an opportunity, or monitor and evaluate current business operations. The research objectives cannot be developed until managers and researchers have agreed on the actual

research objectives

The goals to be achieved by conducting research.

deliverables

The term used often in consulting to describe research objectives to a research client.

business “problem” that will be addressed by the research. Thus, they set out to “discover” this problem through a series of interviews and through a document called a research proposal.

It should be noted that this process is oriented more toward *discovery* than *confirmation* or *justification*. Managers and researchers alike may not have a clear-cut understanding of the situation at the outset of the research process. Managers may only be able to list symptoms that could indicate a problem. For example, employee turnover is increasing, but management may not know the exact nature of the problem. Thus, the problem statement often is made only in general terms; what is to be investigated is not yet specifically identified.

■ DEFINING THE MANAGERIAL DECISION SITUATION

In business research, the adage “a problem well defined is a problem half solved” is worth remembering. Similarly, Albert Einstein noted that “the formulation of a problem is often more essential than its solution.”⁹ These phrases emphasize that an orderly definition of the research problem provides direction to the investigation. Careful attention to problem definition allows the researcher to set the proper research objectives. When the purpose of the research is clear, the chances of collecting the necessary and relevant information, and not collecting surplus information, will be much greater.

Managers often are more concerned with finding the right answer rather than asking the right question. They also want one solution quickly, rather than having to spend time considering many possible solutions. However, properly defining a problem can be more difficult than actually solving it. In business research, if data are collected before the nature of the problem is carefully thought out, they probably will not yield useful information.

Thus, defining the decision situation must precede the research objectives. Frequently the researcher will not be involved until the management team has discovered that some information about a particular aspect of the business is needed. Even at this point the exact nature of the situation may be poorly defined. Once a problem area has been discovered, the researcher and management together can begin the process of precisely defining it.

Much too often research is conducted without a clear definition of the objectives. Researchers forget that the best place to begin a research project is at the end. In other words, knowing what is to be accomplished determines the research process. An error or omission in specifying objectives is likely to be a costly mistake that cannot be corrected in later stages of the research process.

The library contains a wealth of information. Studies forming a literature review can be found in the library.



■ EXPLORATORY RESEARCH

Exploratory research can be used to help identify and clarify the decisions that need to be made. These preliminary research activities can narrow the scope of the research topic and help transform ambiguous problems into well-defined ones that yield specific research objectives. By investigating any existing studies on the subject, talking with knowledgeable individuals, and informally investigating the situation, the researcher can progressively sharpen the focus of the research. After such exploration, the researcher should know exactly which data to collect during the formal phases of the project and how to conduct the project. Exhibit 4.6 indicates that managers and researchers must decide whether to use one or more exploratory research techniques. As Exhibit 4.6 indicates, this stage is optional.

The business researcher can employ techniques from four basic categories to obtain insights and gain a clearer idea of the problem: previous research, pilot studies, case studies, and experience surveys. These are discussed in detail in later chapters. This section will briefly discuss previous research and focus group interviews, the most popular type of pilot study.

Previous Research

As a general rule, researchers should first investigate previous research to see whether or not others may have already addressed similar research problems. Initially, internal research reports should be searched within the company's archives. In addition, some firms specialize in providing various types of research reports, such as economic forecasts. The *Census of Population* and the *Survey of Current Business* are each examples of previous research conducted by an outside source.

Previous research may also exist in the public domain. The first place researchers will likely look today is online. The Internet and modern electronic search engines available through most university libraries have made literature reviews simpler and faster to conduct. A **literature review** is a directed search of published works, including periodicals and books, that discusses theory and presents empirical results that are relevant to the topic at hand. While a literature survey is common in applied research studies, it is a fundamental requirement of a basic research report.

Suppose, for example, that a bank is interested in determining the best site for additional automated teller machines. A logical first step would be to investigate the factors that bankers in other parts of the country consider important. By reading articles in banking journals, management might quickly discover that the best locations are inside supermarkets located in residential areas where people are young, highly educated, and earning higher-than-average incomes. These data might lead the bank to investigate census information to determine where in the city such people live. Reviewing and building on the work already compiled by others is an economical starting point for most research.

Pilot Studies

Almost all consumers take a test drive before buying a car. A pilot study serves a similar purpose for the researcher. A **pilot study** is a small-scale research project that collects data from respondents similar to those that will be used in the full study. It can serve as a guide for a larger study or examine specific aspects of the research to see if the selected procedures will actually work as intended. Pilot studies are critical in refining survey questions and reducing the risk that the full study will be fatally flawed. This is particularly true for experimental research, which depends critically on valid manipulations of experimental variables.¹⁰ Pilot studies also often are useful in fine-tuning research objectives. Pilot studies are sometimes referred to as pretests. A **pretest** is a very descriptive term indicating a small-scale study in which the results are preliminary and intended only to assist in design of a subsequent study.

Focus group interviews are sometimes used as a pilot study. A **focus group** interview brings together six to twelve people in a loosely structured format. The technique is based on the assumption that individuals are more willing to talk about things when they are able to do so within a group discussion format. Focus group respondents sometimes feed on each other's comments to develop ideas that would be difficult to express in a different interview format.

For example, suppose a consultant is hired by Carrefour to research the way consumers react to sales promotions. Carrefour began in France over 50 years ago and pioneered the discount hypermarket format. Carrefour is now the second largest retailer in the world (behind Walmart), operating nearly 11,000 stores in 29 countries. Specifically, the researcher is asked to help Carrefour executives decide whether or not the size of promotional discounts should vary with national culture. In other words, the basic research question is whether or not culture influences consumer perceptions of sales promotions.¹¹ A pretest may be needed to examine whether or not differences in currency might interfere with these perceptions, or whether or not the different terms that refer to promotions and discounts can be translated into the languages of each culture. For example, is a discount expressed in Korean won interpreted the same way as a discount expressed in euros? Each euro equals about \$1.28, whereas a single dollar is worth about 1,380 won.¹²

Exploratory research need not always follow a structured design. Because the purpose of exploratory research is to gain insights and discover new ideas, researchers may use considerable creativity and flexibility. Some companies perform exploratory research routinely as part of environmental scanning. If the conclusions made during this stage suggest business opportunities, the researcher is in a position to begin planning a formal, quantitative research project.

literature review

A directed search of published works, including periodicals and books, that discusses theory and presents empirical results that are relevant to the topic at hand.

pilot study

A small-scale research project that collects data from respondents similar to those to be used in the full study.

pretest

A small-scale study in which the results are only preliminary and intended only to assist in design of a subsequent study.

focus group

A small group discussion about some research topic led by a moderator who guides discussion among the participants.

■ STATING RESEARCH OBJECTIVES

After identifying and clarifying the problem, with or without exploratory research, the researcher must formally state the research objectives. This statement delineates the type of research that is needed and what intelligence may result that would allow the decision maker to make informed choices. The statement of research objectives culminates the process of clarifying the managerial decision into something actionable.

A written decision statement expresses the business situation to the researcher and makes sure that managers and researchers are on the same page. The research objectives try to directly address the decision statement or statements, as the case may be. As such, the research objectives represent a contract of sorts that commits the researcher to producing the needed research. This is why they are expressed as *deliverables* in applied business research. These research objectives drive the rest of the research process. Indeed, before proceeding, the researcher and managers must agree that the objectives are appropriate and will produce relevant information.

■ LINKING DECISION STATEMENTS, OBJECTIVES, AND HYPOTHESES

In Chapter 3 we discussed the role of theory and research hypotheses. Our hypotheses should be logically derived from and linked to our research objectives. For example, using our opening vignette as an example, the researcher may use theoretical reasoning to develop the following hypothesis:

H1: The more hours per week a prospective student works, the more favorable the attitude toward online MBA class offerings.

Exhibit 4.7 illustrates how decision statements are linked to research objectives, which are linked to research hypotheses. Although the first two objectives each have one hypothesis, notice that the third has two. In reality, most research projects will involve more than one research objective, and each of these may often involve more than one hypothesis. Think about how you might go about trying to test the hypothesis listed in Exhibit 4.7.

EXHIBIT 4.7
Example Decision
Statements, Research
Objectives, and Research
Hypotheses

Decision Statement	Research Objectives	Hypotheses
What should be the retail price for product X?	Forecast sales for product X at three different prices.	Sales will be higher at \$5.00 than at \$4.00 or at \$6.99.
In what ways can we improve our service quality?	Identify the top factors that contribute to customers' perceptions.	Cleanliness is related positively to customers' service quality service perceptions. Crowding is related negatively to customers' service quality perceptions.
Should we invest in a training program to reduce role conflict among our employees?	Determine how much role conflict influences employee job satisfaction.	Role conflict is related positively to job satisfaction.

Planning the Research Design

After the researcher has formulated the research problem, he or she must develop the research design as part of the research design stage. A **research design** is a master plan that specifies the methods and procedures for collecting and analyzing the needed information. A research design provides a framework or plan of action for the research. Objectives of the study determined during the early stages of research are included in the design to ensure that the information collected is appropriate for solving the problem. The researcher also must determine the sources of information, the design technique (survey or experiment, for example), the sampling methodology, and the schedule and cost of the research.

research design

A master plan that specifies the methods and procedures for collecting and analyzing the needed information.

■ SELECTION OF THE BASIC RESEARCH METHOD

Here again, the researcher must make a decision. Exhibit 4.6 shows four basic design techniques for descriptive and causal research: surveys, experiments, secondary data, and observation. The objectives of the study, the available data sources, the urgency of the decision, and the cost of obtaining the data will determine which method should be chosen. The managerial aspects of selecting the research design will be considered later.

In business research, the most common method of generating primary data is the survey. Most people have seen the results of political surveys by Gallup or Harris Online, and some have been respondents (members of a sample who supply answers) to research questionnaires. A **survey** is a research technique in which a sample is interviewed in some form or the behavior of respondents is observed and described in some way. The term *surveyor* is most often reserved for civil engineers who describe some piece of property using a transit. Similarly, business researchers describe some group of interest (such as executives, employees, customers, or competitors) using a questionnaire. The task of writing a list of questions and designing the format of the printed or written questionnaire is an essential aspect of the development of a survey research design.

survey

A research technique in which a sample is interviewed in some form or the behavior of respondents is observed and described in some way.

Research investigators may choose to contact respondents by telephone or mail, on the Internet, or in person. An advertiser spending \$3 million for 30 seconds of commercial time during the Super Bowl may telephone people to quickly gather information concerning their responses to the advertising. The economic development director for a city trying to determine the most important factors in attracting new businesses might choose a mail questionnaire because the appropriate executives are hard to reach by telephone. A manufacturer of a birth control device for men might determine the need for a versatile survey method wherein an interviewer can ask a variety of personal questions in a flexible format. While personal interviews are expensive, they are valuable because investigators can use visual aids and supplement the interviews with observations. Each of these survey methods has advantages and disadvantages. A researcher's task is to find the most appropriate way to collect the needed information in a particular situation.

The objective of many research projects is merely to record what can be observed—for example, the number of automobiles that pass by a proposed site for a gas station. This can be mechanically recorded or observed by humans. Research personnel known as mystery shoppers may act as customers to observe actions of sales personnel or do comparative shopping to learn prices at competing outlets. A mystery shopper is paid to pretend to be a customer and gather data about the way employees behave and the way they are treated in general. How often are store policies followed? Are they treated courteously? Mystery shoppers can be valuable sources for observational data.

The main advantage of the observation technique is that it records behavior without relying on reports from respondents. Observational data are often collected unobtrusively and passively without a respondent's direct participation. For instance, Nielsen Media Research uses a "people meter" attached to television sets to record the programs being watched by each household member. This eliminates the possible bias of respondents stating that they watched the president's State of the Union address rather than *Gossip Girl* on another station.

Observation is more complex than mere "nose counting," and the task is more difficult than the inexperienced researcher would imagine. While observation eliminates potential bias from interviewer interaction, several things of interest, such as attitudes, opinions, motivations, and other intangible states of mind, simply cannot be observed.

■ THE "BEST" RESEARCH DESIGN

It is argued that there is no single best research design. As such, the researcher often has several alternatives that can accomplish the stated research objectives. Consider the researcher who must forecast sales for the upcoming year. Some commonly used forecasting methods are surveying executive opinion, collecting sales force composite opinions, surveying user expectations, projecting trends, and analyzing environmental factors. Any one of these may yield a reliable forecast.

The ability to select the most appropriate research design develops with experience. Inexperienced researchers often jump to the conclusion that a survey methodology is usually the best design because they are most comfortable with this method. When Chicago's Museum of Science

RESEARCH SNAPSHOT



Rolling Rock

Making a mark in the U.S. beer market can be difficult. American consumers tend to favor milder beers at lower price points. Some argue that most beers taste very similar. Taste tests do reveal that similarly positioned beers do taste very much the same. However, the taste rankings do not correspond to market share. For instance, Stroh's fared very well in the taste tests, but it is hardly a market leader. Rolling Rock rated 12th out of 12 beers tasted. Tasters said it tasted a bit like canned corn. Clearly, there is something more to a successful beer than taste.

For many years Rolling Rock beer was a regional brand in western Pennsylvania. Its signature package was a longneck green bottle with a white painted label featuring icons such as a horse head, a steeplechase, the number "33," and a legend about the beer being brought to you "from the glass-lined tanks of Old Latrobe." The brand, now sold by Labatt USA, expanded nationally during the 1980s by focusing on core consumers who purchased specialty beers for on-premise consumption and who were willing to pay higher prices than for national brands such as Budweiser.

As years went by, packaging options expanded to include bottles with ordinary paper labels for take-home consumption, often packaged in 12-packs. In the mid-1990s, in response to a competitive explosion from microbrews, Rolling Rock offered a number of line extensions, such as Rock Bock and amber Rock Ice. They failed. Sales stagnated. In New York and other crucial markets, price reductions to the level of Budweiser and Miller became inhibiting aspects of its marketing program. Business executives held the view that the longneck painted bottle was the heart of the brand. However, earlier efforts to develop cheaper imitations of the painted-label look had not achieved success.



Rolling Rock executives decided to conduct a massive consumer study, recruiting consumers at shopping malls and other venues to view "live" shelf sets of beer—not just specialty beer, but beer at every price range from subpremiums and up. Consumers given money to spend in the form of chips were exposed to "old-bundle" packages (the old graphics and the paper-label stubbies) and "new-bundle" packages (two new graphics approaches, including the one ultimately selected, and painted-label longnecks) at a variety of price points and asked to allocate chips to their next ten purchases. Some were even invited to take the "new-bundle" packages home with them for follow-up research.

As the business executives had hoped, the results did not leave any room for interpretation: Not only did the new packages meet with consumers' strong approval, but consumers consistently indicated that they would be willing to pay more for the brand in those packages. In fact, not only were they willing to pay more; they *expected* to pay more, particularly among consumers already familiar with the Rock. In three regions, the Northeast, Southeast, and West, purchase-intent among users increased dramatically both at prices 20 cents higher per 12-pack and at prices 40 cents higher per 12-pack. The increase in purchase intent was milder in the Midwest, but there Rock already commanded a solid premium over Bud and other premium beers. The sole exception to that trend was in the brand's core markets in Pennsylvania and Ohio, where Rock has never entirely escaped its shot-and-a-beer origins, but even there, purchase intent declined by only 2 percent at each of the higher prices.

Sources: Gerry Khermouch, "Sticking Their Neck Out," *BrandWeek* (November 9, 1998) 25–34, © 2006 VNU Business Media, Inc. Used with permission from Brandweek. © 1998–1999 VNU Business Media Inc.; "Which Brew for You?" *Consumer Reports* (August 2001), 10–17.

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TO THE POINT

You cannot put the same shoe on every foot.

—Publius Syrus

and Industry wanted to determine the relative popularity of its exhibits, it could have conducted a survey. Instead, a creative researcher familiar with other research designs suggested a far less expensive alternative: an unobtrusive observation technique. The researcher suggested that the museum merely keep track of the frequency with which the floor tiles in front of the various exhibits had to be replaced, indicating where the heaviest traffic occurred. When this was done, the museum found that the chick-hatching exhibit was the most popular. This method provided the same results as a survey but at a much lower cost. Take a look at the research design used by Rolling Rock illustrated in the Snapshot above.

Sampling

Although the sampling plan is outlined in the research design, the sampling stage is a distinct phase of the research process. For convenience, however, we will treat the sample planning and the actual sample generation processes together in this section.

If you take your first bite of shrimp po-boy and conclude that it needs Tabasco, you have just conducted a sample. **Sampling** involves any procedure that draws conclusions based on measurements of a portion of the population. In other words, a sample is a subset from a larger population. If certain statistical procedures are followed, a researcher need not select every item

sampling

Involves any procedure that draws conclusions based on measurements of a portion of the population.

in a population because the results of a good sample should have the same characteristics as the population as a whole. Of course, when errors are made, samples do not give reliable estimates of the population.

A famous example of error due to sampling is the 1936 *Literary Digest* fiasco. The magazine conducted a survey and predicted that Republican Alf Landon would win over Democrat Franklin D. Roosevelt by a landslide in that year's presidential election. This prediction was wrong—and the error was due to sample selection. The post-mortems showed that *Literary Digest* had sampled its readers and names drawn from telephone books and auto registrations. In 1936, not everyone had a telephone or a car; thus the sample was biased toward people with means. In reality, Roosevelt received over 60 percent of the popular vote.

In 2004, early exit polls led many to believe that John Kerry would win the U.S. presidential election.¹³ The exit polls were performed early on election day and done mostly in highly urban areas in the Northeast, areas that are predominantly Democratic. The resulting sample of voters responding to the early exit polls did not represent the entire U.S. population, and Kerry lost to Bush by over 3 million votes, or about 3 percent of all votes cast. Thus, the accuracy of predictions from research depends on getting a sample that really matches the population.

The first sampling question to ask is “Who is to be sampled?” The answer to this primary question requires the identification of a *target population*. Who do we want the sample to reflect? Defining this population and determining the sampling units may not be so easy. If, for example, a savings and loan association surveys people who already have accounts for answers to image questions, the selected sampling units may represent *current* customers but will not represent *potential* customers. Specifying the target population is a crucial aspect of the sampling plan.

The next sampling issue concerns sample size. How big should the sample be? Although management may wish to examine every potential buyer of a product or service, doing so may be unnecessary as well as unrealistic. Other things equal, larger samples are more precise than smaller ones. However, proper probability sampling can allow a small proportion of the total population to give a reliable measure of the whole. A later discussion will explain how large a sample must be in order to be truly representative of the universe or population. Essentially, this is a question of how much variance exists in the population.

The final sampling decision is how to select the sampling units. Simple random sampling may be the best known type, in which every unit in the population has an equal and known chance of being selected. However, this is only one type of sampling. For example, if members of the population are found in close geographical clusters, a cluster sampling procedure (one that selects area clusters rather than individual units in the population) will reduce costs. Rather than selecting 1,000 individuals throughout the United States, it may be more economical to first select 25 counties and then sample within those counties. This will substantially reduce travel, hiring, and training costs. In determining the appropriate sampling plan, the researcher will have to select the most appropriate sampling procedure for meeting the established study objectives.

Gathering Data

The data gathering stage begins once the sampling plan has been formalized. Data gathering is the process of gathering or collecting information. Data may be gathered by human observers or interviewers, or they may be recorded by machines as in the case of scanner data and Web-based surveys.

Obviously, the many research techniques involve many methods of gathering data. Surveys require direct participation by research respondents. This may involve filling out a questionnaire or interacting with an interviewer. In this sense, they are obtrusive. **Unobtrusive methods** of data gathering are those in which the subjects do not have to be disturbed for data to be collected. They may even be unaware that research is going on at all. For instance, a simple count of motorists driving past a proposed franchising location is one kind of data gathering method. However the data are collected, it is important to minimize errors in the process. For example, the data gathering should be consistent in all geographical areas. If an interviewer phrases questions incorrectly or records a respondent's statements inaccurately (not verbatim), major data collection errors will result.

unobtrusive methods

Methods in which research respondents do not have to be disturbed for data to be gathered.

Processing and Analyzing Data

■ EDITING AND CODING

After the fieldwork has been completed, the data must be converted into a format that will answer the manager's questions. This is part of the data processing and analysis stage. Here, the information content will be mined from the raw data. Data processing generally begins with editing and coding the data. Editing involves checking the data collection forms for omissions, legibility, and consistency in classification. The editing process corrects problems such as interviewer errors (an answer recorded on the wrong portion of a questionnaire, for example) before the data are transferred to the computer.

Before data can be tabulated, meaningful categories and character symbols must be established for groups of responses. The rules for interpreting, categorizing, recording, and transferring the data to the data storage media are called codes. This coding process facilitates computer or hand tabulation. If computer analysis is to be used, the data are entered into the computer and verified. Computer-assisted (online) interviewing is an example of the impact of technological change on the research process. Telephone interviewers, seated at computer terminals, read survey questions displayed on the monitor. The interviewer asks the questions and then types in the respondents' answers. Thus, answers are collected and processed into the computer at the same time, eliminating intermediate steps that could introduce errors.

■ DATA ANALYSIS

data analysis

The application of reasoning to understand the data that have been gathered.

Data analysis is the application of reasoning to understand the data that have been gathered. In its simplest form, analysis may involve determining consistent patterns and summarizing the relevant details revealed in the investigation. The appropriate analytical technique for data analysis will be determined by management's information requirements, the characteristics of the research design, and the nature of the data gathered. Statistical analysis may range from portraying a simple frequency distribution to more complex multivariate analyses approaches, such as multiple regression. Later chapters will discuss three general categories of statistical analysis: univariate analysis, bivariate analysis, and multivariate analysis.

Drawing Conclusions and Preparing a Report

One of the most important jobs that a researcher performs is communicating the research results. This is the final stage of the research project, but it is far from the least important. The conclusions and report preparation stage consists of interpreting the research results, describing the implications, and drawing the appropriate conclusions for managerial decisions. These conclusions should fulfill the deliverables promised in the research proposal. In addition, it's important that the researcher consider the varying abilities of people to understand the research results. The report shouldn't be written the same way to a group of Ph.D.'s as it would be to a group of line managers.

All too many applied business research reports are overly complicated statements of technical aspects and sophisticated research methods. Frequently, management is not interested in detailed reporting of the research design and statistical findings, but wishes only a summary of the findings. If the findings of the research remain unread on the manager's desk, the study will have been useless. The importance of effective communication cannot be overemphasized. Research is only as good as its applications.

Now that we have outlined the research process, note that the order of topics in this book follows the flowchart of the research process presented in Exhibit 4.4. Keep this flowchart in mind while reading later chapters.



The Research Program Strategy

Our discussion of the business research process began with the assumption that the researcher wished to collect data to achieve a specific organizational objective. When the researcher has only one or a small number of research objectives that can be addressed in a single study, that study is